2023 Ch H1 Q1

Section: Chemical Changes and Structure

Topic: Structure and Bonding

Question Summary

Which of the following compounds has the least ionic character? A: sodium iodide (NaI), B: sodium fluoride (NaF), C: potassium iodide (KI), D: potassium fluoride (KF).

Worked Solution

Ionic character increases with the difference in electronegativity between the bonded atoms. Fluorine is far more electronegative than iodine, so fluorides (NaF, KF) are more ionic than iodides (NaI, KI).

Between the remaining iodides, consider Fajans' rules: a smaller, more highly charged cation polarises the large iodide anion more strongly, increasing covalent character (i.e. reducing ionic character). Sodium ions (Na^+) are smaller than potassium ions (K^+), so NaI is more covalent (less ionic) than KI.

Final Answer

A — Sodium iodide (NaI)

Revision Tips

• Bigger ΔEN → more ionic; smaller ΔEN → more covalent.
• Larger anions (like I^-) and smaller cations (like Na^+) increase polarisation → more covalent character.
• Order of ionic character here: KF ≈ NaF > KI > NaI (least).